

What are we trying to find out using this project?

We tried to examine:

- Infrastructure and Policy Impact: How government incentives, tax credits, and local utility investments correlate with regional adoption rates across counties and postal codes.
- Economic Determinants of Adoption: How household purchasing power (reflected via MSRP ranges and local income/census tract demographics) influences the transition from conventional vehicles to Battery Electric Vehicles (BEVs).
- Technological Diffusion: The shifting consumer preference away from short-range hybrid models toward long-range BEVs as battery technology and vehicle availability mature over time.

What we performed in the steps:

Dataset Dimensions: Confirmed a total size of 135,038 rows and 17 columns.

Data Types & Memory: Showed that the DataFrame consists of 5 float64 columns, 2 int64 columns, and 10 object (string) columns, consuming roughly 17.5 MB of memory.

Sample Data Preview: Provided visibility into core attributes such as VIN snippets, Geographic locations (County, City, State, Postal Code, Census Tract), Vehicle specifications (Make, Model, Model Year, EV Type, CAFV Eligibility, Electric Range, Base MSRP), and administrative identifiers (DOL Vehicle ID, Legislative District).

A precise audit of data gaps across the columns, highlighting that fields like Legislative District (312 missing), Model (249 missing), Vehicle Location (10 missing), and minor geographical/utility fields (8 missing each) require attention or awareness during modeling and grouping.

Model Year: Ranged from 1997 to 2024, with a mean year of approximately 2020 (median 2021), indicating a heavily recent-skewed, modern EV population.

Electric Range: Showed a mean range of 74.59 miles, with a median of 21 miles and a maximum range of 337 miles (reflecting a mix of Plug-in Hybrid Electric Vehicles with low electric-only range and long-range Battery Electric Vehicles).

Base MSRP: Highlighted a maximum base manufacturer's suggested retail price of \$845,000, though the 25th, 50th, and 75th percentiles sat at \$0.00, indicating that many vehicle records do not report MSRP data in this specific registry.

Conclusion based on the 7 steps that we performed

- Exponential Adoption Growth: The market has experienced a massive, non-linear acceleration in EV registrations, particularly surging from the mid-2010s onward to peak modern adoption levels.

- Shift Toward Long-Range Battery EVs: While historical models and plug-in hybrids kept average electric ranges relatively low (with a median of 21 miles), the availability of models reaching up to 337 miles highlights a decisive industry shift toward high-capacity Battery Electric Vehicles (BEVs).
- High Market Variance & Premium Segment Presence: The data reflects a diverse pricing structure, ranging from standard consumer vehicles up to luxury/high-performance entries with MSRPs reaching \$845,000.
- Data Integrity & Standardization Readiness: Rigorous auditing and cleaning (identifying minor missing values in auxiliary fields like legislative districts and standardizing text whitespace) confirmed that the dataset is robust, highly structured, and fully prepared for advanced predictive modeling or spatial-temporal economic analysis.